

✦ S I L L A G E

Perfume Recommendation System

Powered by TF-IDF · K-Means Clustering · Association Rule Mining

THE PROBLEM

Finding the right fragrance is overwhelming.



1000s of Options

Perfume databases contain thousands of fragrances with complex note profiles.



No Guidance

Buyers struggle to communicate what they like — notes, accords, families.



Costly Mistakes

Purchasing the wrong scent is expensive with no easy way to discover alternatives.

Sillage — An AI-Powered Perfume Recommender

- **User Input** · Preferred & disliked notes, accords, gender, rating threshold
- **Smart Filtering** · TF-IDF vectors rank perfumes by scent-note similarity
- **Clustering** · K-Means groups perfumes into scent families for exploration
- **Association Rules** Apriori mining surfaces 'you may also like' suggestions
- **Streamlit UI** · Luxury-styled web app with charts, filters & live results

Data Mining Pipeline

01

Data Collection

perfumes.xlsx — notes, accords, ratings & gender

02

Preprocessing

Tokenisation, stop-word removal, TF-IDF
vectorisation

03

PCA

Dimensionality reduction to visualise scent clusters
in 2D

04

K-Means Clustering

Unsupervised grouping of perfumes into scent
families

05

Association Rules

Apriori mining — frequent note co-occurrences for
'also like'

06

Cosine Similarity

User vector matched against TF-IDF matrix; top-N
returned

07

Recommender System

Combines all signals to deliver personalised ranked
results

KEY FEATURES

What the App Delivers



Smart Recommendations

Top-N perfumes ranked by cosine similarity to your taste profile



Association Suggestions

"Also You May Like" powered by Apriori rule mining



Cluster Explorer

Browse fragrance families discovered by K-Means



Dataset Insights

Interactive charts: note distributions, rating histograms



Note Blacklisting

Exclude disliked notes from all recommendations



Streamlit Web UI

Luxury-styled, interactive and easy to use

✦ S I L L A G E

Thank You

O U R T E A M



Islam Sald Aldeen



Rawan Issa



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